Smart watering controller — PCB design review

Board: `watering.kicad_pcb` / `watering.kicad_sch` **Reviewed:** 28 August 2026

1. Board summary (measured from the files)

Property	Value
Outline	76.45 × 50.80 mm, rounded corners
Layers	2 (F.Cu signal, B.Cu power/ground), 1.6 mm FR-4
Copper	1 oz both sides
Surface finish set in stackup	HAL SnPb
Footprints	50 (45 unique reference designators)
Track segments	442 — 362 on F.Cu (891.4 mm), 80 on B.Cu (206.8 mm)
Track widths in use	0.25 / 0.50 / 0.75 / 1.00 mm
Vias	117, all 0.7 mm pad / 0.3 mm drill
Copper zones	5 filled (GND on both layers, plus +9V, +4V2, BATT_GND, BOOST_9V_SW on F.Cu) + 1 keepout
Minimum track-to-track clearance	0.200 mm
B.Cu ground fill	one continuous region, 3313 mm² (~85 % of the board)
F.Cu ground fill	12 separate islands (1859 mm² main + 11 from 2 to 80 mm²)
GND stitching vias	74

System architecture as built

- 18650 cell(s) in parallel → `BAT_J1` → low-side protection (`S-82A1A` + `CSD83325L` + 5 mΩ shunt) → `BATT_GND`
- Battery positive leaves on `/AEM_PWR_STO` to the AEM breakout and returns as `+4V2`
- `+3V3` arrives from the AEM breakout LDO on `AEM_CONN_PWR1.2`
- `TPS61089` boost `+4V2 → +9V` for the pump, enabled by `/PUMP_POWER_CTRL`
- `LDL1117S50R` `+9V → +5V` for the `MPX5010DP` pressure sensor
- `BL54L15` module (nRF54L15) with `CH340E` USB-UART bridge and USB-C

2. What is already good

- Block placement is coherent: battery and protection top-left, harvester connectors on the left edge, boost and pump path down the middle-left, MCU top-right, USB and CH340E on the right.
- Power nets are widened appropriately. At 1 oz outer copper and a 10 °C rise (IPC-2221), 1.00 mm gives roughly 2.4 A and 0.75 mm roughly 1.9 A, against a worst case well under 1 A. No current-capacity concerns anywhere on the board.
- All four module GND pads have a via within 1.1–1.2 mm.
- B.Cu really is a single continuous ground region covering about 85 % of the board.
- All four mounting holes are `MountingHole_2.5mm_Pad_Via` on GND, so they contribute stitching.
- Clearance is a consistent 0.200 mm — comfortable for any 2-layer fab.
- None of the twelve F.Cu ground islands is orphaned; each has at least one via to the bottom plane.
- USB-C CC pull-downs (2 × 5.1 kΩ) are correct for a sink, and the `USBLC6-2SC6` is correctly placed between the receptacle and the bridge.

3. Circuit-level issues (fix before ordering)

These matter more than any routing problem.

3.1 The MPX5010DP will overdrive the ADC pin

At a 5 V supply the MPX5010DP outputs 0.2 V to 4.7 V over its 0–10 kPa range. `/PRESSURE_OUT` runs directly from `PRESSURE_U1.1` to `MCU_IC1.32` (P1.14/AIN7) on a 3.3 V module. Absolute maximum on that pin is VDD + 0.3 V. A 2.5 W pump will pass 10 kPa easily, so this is not a corner case.

Fix: resistive divider at the connector. 10 kΩ / 18 kΩ maps 4.7 V to 3.02 V. The divider's Thévenin resistance (6.4 kΩ) doubles as the series protection resistor; add 10 nF across the lower leg for an anti-alias filter.

3.2 The boost enable floats

`/PUMP_POWER_CTRL` connects only to `TPS61089.7` (EN), `MCU_IC1.23` and a pass-through pin on the AEM data connector. There is no pull-down. During reset, during SWD programming, and before firmware configures the pin, EN is high-impedance — the pump can run when it should not.

Fix: 100 kΩ from EN to GND, physically at the IC.

3.3 The CH340E is permanently powered from the battery

Its VCC (pin 7) and V3 (pin 10) sit on `+3V3`, which comes from the AEM board — not from VBUS. The bridge therefore draws current continuously on a solar-harvested device even with no USB cable attached.

Fix: either derive its 3.3 V from VBUS with a small LDO, or put a high-side switch on its `+3V3` supply gated by VBUS presence. Add ~1 kΩ in series on `/UART00_TX`, `/UART00_RX`, `/UART00_CTS` and `/UART00_RTS` either way, so the MCU cannot push current into an unpowered CH340E's ESD structures.

A `TPS22917DBVR` works here (see §5.7), but **do not drive its ON pin directly from VBUS** — the pin's absolute maximum is 6 V and USB transients exceed that. Use a divider, and size it knowing the 750 kΩ smart pulldown is still connected until ON first crosses VIH. With 1 MΩ / 1 MΩ the lower leg becomes 1 MΩ ‖ 750 kΩ and you only reach about 1.5 V before the pulldown releases. 470 kΩ / 470 kΩ gives roughly 1.9 V under that load and draws about 5 µA — from VBUS, not from the battery.

3.4 The LDL1117 has no local input capacitor

Its VIN pad is at (120.20, 113.02). The nearest `+9V` capacitor is 16.2 mm away.

Fix: 1 µF + 100 nF right at the pin. Also pull the 10 µF output cap in from its current 4.6 mm.

3.5 No bulk capacitance on VBUS

The `USBLC6-2SC6` gives the TVS clamp, but nothing else is on the net. USB expects ≥ 1 µF at the receptacle.

Fix: 4.7–10 µF plus 100 nF near `USB_J1`.

3.6 Nothing local at the pump connector

`PUMP_U1.2` is 19.8 mm of 1 mm trace from the nearest `+9V` capacitor. A BLDC driver commutating at the end of that inductance will ring.

Fix: 10–22 µF plus 100 nF at the connector, and a TVS.

3.7 Smaller items

- `LDL1117S50R` pad 2 is unconnected. Tie it to VOUT (it is the same node as the tab internally).
- `/AEM_ESRC1` and `/AEM_ESRC_GND` are single-pad nets — the connector pins go nowhere on this board. Confirm that is intentional and that the solar panel really lands on the harvester breakout.
- Five duplicate reference designators: `22uF`, `100nF`, `470nF`, `5.1kO` and `10uF` each appear twice, because the refdes field holds the value. Annotate before generating a BOM or pick-and-place file.
- Consider ENIG instead of the HAL SnPb currently set in the stackup. The `TPS61089` is a 0.5 mm pitch VQFN-HR with 0.25 mm pads and the `CH340E` is MSOP-10; HASL's uneven surface is marginal at that pitch.
- `+3V3` has 108 mm of routing from a connector on the far left, with a single 10 µF at 3.3 mm from the module's VDD pin as its only local reservoir, shared with the CH340E's current bursts. Add 100 nF right at pin 26 and give the CH340E more than its current 2 × 470 nF.

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4. Layout and routing

4.1 The antenna region — highest priority

The keepout zone spans x 124.46–143.51, y 76.20–81.15. The BL54L15 body spans x 135.11–145.11, y 76.21–90.21. **The keepout stops 1.6 mm short of the module's right edge**, and there is ground pour on both layers in that strip, directly under the antenna end.

Copper found in and beside the antenna region:

Net	Geometry	Problem
`/SWDIO`	(144.61, 81.41) → (144.00, 81.20) → (144.00, 77.90) → (145.02, 76.83) → (162.2, 76.83)	Runs under the module body inside the antenna region, then 17 mm along the top edge at 0.50 mm copper-to-edge
`/SOIL1_VCC`	y = 77.5 from x = 147.4 to 149.7	Immediately beside the antenna end
`/SOIL2_VCC`	y = 81.3, 10 mm long	Same
`/SOIL3_VCC`	diagonal through (147.2, 80.7)	Same
GND vias	(145.7, 78.0), (147.2, 79.0), (149.1, 80.5)	Grounded copper adjacent to the radiating

		element
`SENSE_U2`	(148.88, 78.70)	Through-hole connector with an off-board cable, 3.8 mm from the antenna end at the same y

Cables running past a PCB antenna at that distance detune it and become unintentional radiators.

Fix: take the exact ground-clear dimensions from the Ezurio BL54L15 integration guide and enforce them on every layer with margin on all four sides. The cleanest solution is to rotate the assembly so the antenna faces a board corner with nothing beyond it, and move the three soil connectors to the bottom or left edge. Also consider where the 18650 pack physically sits — a metal cell pack near the antenna will do the same thing.

4.2 The ground plane is cut where it matters most

Six nets pass under the module footprint on B.Cu: `/I2C_SDA`, `/I2C_SDC`, `/SOIL1_OUT`, `/SOIL2_OUT`, `/SOIL3_OUT`, `/PUMP_PWM`.

`/PUMP_PWM` in particular runs on B.Cu from (142.47, 98.27) up to (142.47, 88.91), terminating directly beneath the module. A switching motor-drive signal under the RF ground reference is the one route to avoid above all others.

Separately, `/AEM_IRQ`, `/AEM_STO` and `/AEM_LOAD` run as three parallel B.Cu traces at y = 89.6 / 90.3 / 91.0 from x ≈ 110 to x ≈ 133. That is a slot roughly 2.5 mm wide and 22 mm long in the bottom plane, immediately below the MCU. Every F.Cu signal crossing it has to detour its return current around the ends.

4.3 Battery current runs on the ground layer

`/AEM_PWR_STO` is 1.0 mm × 24.9 mm on B.Cu, from `BAT_J1` to `AEM_CONN_PWR1.4`. It returns as `+4V2` on pin 3 and travels 15 mm on F.Cu to the boost input.

The full pump current — roughly 0.7 A drawn from 4.2 V for 2.5 W at 9 V — therefore makes a round trip out to the breakout board through one JST PH pin and back through another, while cutting a second slot down the left side of the plane. Tap `+4V2` locally from the protected battery node on this board if the harvester's topology allows it.

4.4 The boost converter loop is loose

- SW pin (127.52, 103.51) to the inductor pad is **7.16 mm**, and there is a 10.8 × 3.9 mm filled zone on the SW node. That is the highest dV/dt copper on the board and currently the largest single-net pour, 15 mm from a 2.4 GHz receiver. Put the inductor adjacent to the pin and make SW only as large as the current requires.
- Output capacitors sit 3.85 mm (`BOOST_9V_C6`), 5.71 mm (`C7`) and about 7.5 mm (the third 22 µF) from VOUT. On a boost, the **output** loop is the critical high-di/dt path, not the input loop.
- Pads 5 (GND) and 6 (VOUT) are only 0.70 mm apart. Put a 1 µF 0402 straight across them and a 22 µF within 2 mm.
- There is exactly **one GND via within 3 mm** of the IC's GND pad, and it is 2.0 mm away. That pad is both the switching return and the thermal path for a 0.35 × 1.0 mm pad. Add a cluster of 4–6 vias at the pad.
- `/PRESSURE_OUT` runs on B.Cu vertically at x = 130.41 from y = 95.4 to 109.7, about 2 mm from the boost's right edge and directly past the output capacitors. A high-impedance analog signal on the return layer through the noisiest region of the board. Move it.
- Check the inductor ripple with the 1.8 µH part at whatever switching frequency the 301 kΩ FSW resistor sets, and confirm the inductor's saturation rating against the peak, not the average.

4.5 Ground stitching is thin

74 GND vias. Mean distance from any point on the board to the nearest one is 6.1 mm, the 90th percentile is 13.2 mm, and the worst point (around x 150, y 126) is 23.2 mm. About 1400 mm², roughly a third of the board, is more than 6 mm from a stitching via.

For 2.4 GHz you want the perimeter and the area around the module stitched at 5 mm or tighter.

4.6 Minor

- `/PUMP_POWER_CTRL` runs 0.2 mm from `/XL2` near (140.3, 91.1). EN is a static signal so the coupling risk is low, but XL1/XL2 are high-impedance 32.768 kHz nodes and deserve a ground guard with nothing else adjacent.
- The crystal is 6.7 mm from the module and its nearest GND via is 4.6 mm away. Shorten the XL routing and put ground directly under and around the crystal.

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5. Protecting the off-board sensor lines

Every soil sensor, the pressure sensor, the pump and the three buttons leave the board on cables. None of those lines currently has series resistance, filtering or ESD protection.

5.1 What the series resistor is actually for

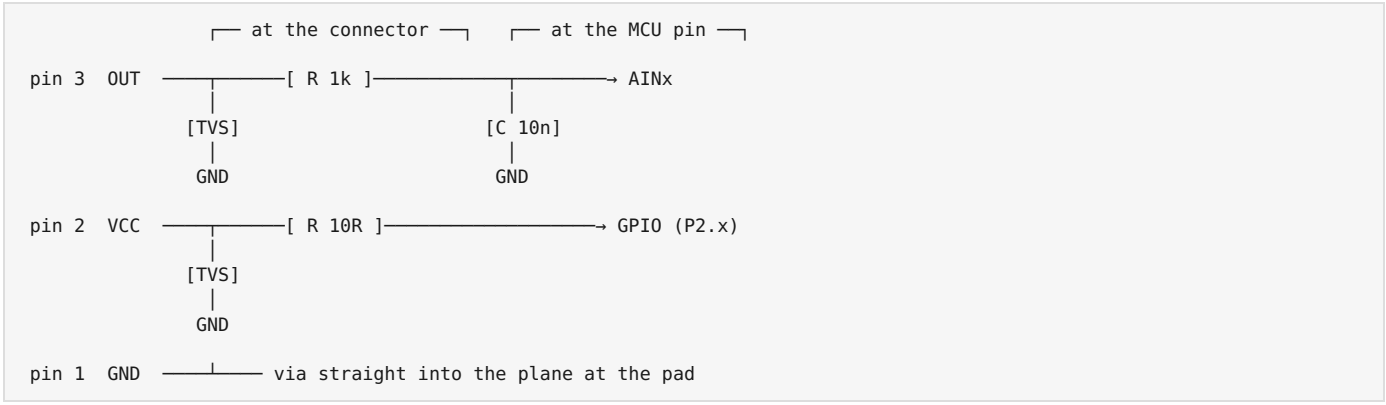
It is **not** primarily short-circuit protection. Ranked by importance:

1. **ESD and surge current limiting.** A metre or more of cable in a garden is a charge collector. When a surge arrives, the MCU's internal clamp diodes conduct to VDD or GND. Those diodes handle a few tens of milliamps, not the tens of amps an ESD event

delivers. The series resistor is what turns the surge into a current the clamp can survive. A TVS placed ahead of the resistor takes the bulk of the energy; the resistor handles whatever gets past it.
2. **Anti-alias and noise filtering.** Together with a small capacitor at the MCU pin it forms an RC low-pass, which matters because the cable will pick up the boost converter and the pump.
3. **A charge reservoir for the SAADC.** The capacitor supplies the sampling capacitor locally instead of forcing it to charge through the cable.
4. **Back-feed limiting.** When a power-gated sensor is off, its output line can still be driven by cable-coupled charge. The resistor limits what flows into the unpowered sensor's protection structures.

Short-circuit protection is a side benefit. The resistance you would need to genuinely current-limit a hard short is far too large for an analog path, so that job belongs elsewhere.

5.2 Topology, per soil sensor channel



Order matters: **TVS first (cable side), then the series resistor, then the filter capacitor.** If the resistor comes before the TVS, the resistor absorbs the surge instead of the diode and may fail open or short.

5.3 Values and why

- `R` **on OUT = 1 kΩ.** nRF SAADC inputs tolerate source impedance in the ~100 kΩ range at a 3 µs acquisition time, so 1 kΩ costs nothing in accuracy. Check the exact acquisition-time versus source-impedance table for the nRF54L15.
- `C` **at the MCU pin = 10 nF** (up to 100 nF if you want a lower corner). 1 kΩ × 10 nF gives a 16 kHz corner and a 10 µs time constant — irrelevant next to the sensor's own settling time.
- `R` **on VCC = 10 Ω or nothing.** At 5 mA this drops 50 mV. Keep it small, because the HW-390 output is ratiometric to its supply and every millivolt of drop shifts the reading. Its job here is inrush limiting into the cable capacitance and the sensor's bypass capacitor, not protection.
- **TVS:** a low-capacitance array with a 3.3 V standoff. The tidiest option is to reuse the `USBLC6-2SC6` already in the BOM — one per connector, pin 5 (VBUS) tied to `+3V3`, channel 1 on the VCC line, channel 2 on the OUT line. Its rail diodes then clamp both lines to the 3.3 V rail, and you avoid adding a BOM line. Alternatives with the same topology: `PESD3V3L4UG`, `SP0503BAHT`, `SRV05-4`.

5.4 Where to place it physically

- Everything goes in the **first 3–5 mm of trace after the connector pads**, on the same layer, before the signal heads anywhere else.
- The **TVS ground pad needs its own via, at the pad.** At ESD rise times a few nanohenries of trace inductance produces well over a hundred volts, which defeats the clamp. A 5 mm trace to the plane makes the part decorative.
- Give the connector's own GND pin a via **at the pad** so cable-borne common-mode current reaches the plane without travelling across the board.
- Do not route the protected (MCU-side) trace alongside the unprotected (cable-side) trace. If they run parallel, the surge couples straight around the protection.
- On the current layout the three sensor connectors are in the top-right beside the antenna, so do this work as part of the re-placement in §4.1 rather than patching it in place.

5.5 Same pattern elsewhere

- **Pressure:** the 10 kΩ / 18 kΩ divider from §3.1 *is* the series element. Add 10 nF across the lower leg and a TVS at the connector.
- **Pump:** `/PUMP_PWM` and `/PUMP_OUT` go to a motor driver. Series resistors (100 Ω–1 kΩ) plus a TVS array. Do not filter `/PUMP_PWM` so heavily that you round off the PWM edges.
- **Buttons:** series 1 kΩ plus 100 nF to GND gives debounce and protection in one, on top of the internal pull-up.
- **SWD:** at minimum a TVS on `/nRST`, since that header gets handled.

5.6 Note on the HW-390 supply plan

Driving the sensors from P2.x in extra-high-drive mode and sequencing them one at a time is a sound approach. Two things to verify:

1. **The drive-strength figure is specified at a particular output voltage.** "At least 14 mA" is normally quoted at VDD minus some drop. That drop — likely 100–150 mV at a 5 mA load — appears directly in the reading, because the HW-390's output scales

with its supply. It is repeatable, so calibrate it out, or budget for a few percent of error. If you want it gone, drive a small P-FET or load switch from the GPIO instead of powering the sensor from the pin: the drop becomes Rds(on) × 5 mA, which is millivolts, and a cable short can no longer reach the MCU pin.

2. **Settling time.** The HW-390's output goes through an RC filter, so allow 100–200 ms after switching VCC on before sampling. Sequencing already gives you that time naturally.

Also confirm the AEM breakout's LDO can supply the burst (MCU in TX, plus CH340E if USB is attached, plus one sensor) and that there is enough bulk capacitance on `+3V3` to cover it.

5.7 Load switch option — TPS22917DBVR

Specs below are from the TI datasheet, SLVSDW8B rev. B, December 2021.

For the three HW-390 supply rails this part is a good fit and removes several of the concerns above at once.

Parameter	Value	Why it matters here
VIN range	1 V to 5.5 V	Fine for 3.3 V. **Not usable on +9V** — see below
RON at 3.6 V	90 mΩ typ, 110 mΩ max at 25 °C, 150 mΩ max to +105 °C	Under 1 mV drop at 5 mA, versus 100–150 mV across a GPIO
IQ (on)	0.5 µA typ, 1.0 µA max to +85 °C	Negligible during a 200 ms sensor read
ISD (off)	10 nA typ, 100 nA max to +85 °C	Three switches idle ≈ 30 nA. Better than a discrete P-FET with a gate divider
ON pin	Smart pulldown, 750 kΩ typ, disconnects once ON is driven ≥ VIH	Defaults off. No floating-enable problem at power-up
ON pin leakage	±10 nA max	Standard-drive GPIO on any port is enough
QOD	Internal ~150 Ω, or external RQOD, or disabled by leaving the pin open	Actively discharges the sensor rail and its cable when off
Reverse current blocking	Always on; VIN reverse leakage −1 µA max when VOUT > VIN	A bare P-FET's body diode would feed the 3.3 V rail instead
Package	6-pin SOT-23 (DBV), 2.90 × 1.60 mm	Hand-solderable, visible joints

What this buys you. The ratiometric error from §5.6 disappears, because the sensor sees the full rail. A cable short can no longer reach an MCU pin. Extra-high drive is no longer needed at all, which frees the design from the P2 constraint. And the smart pulldown means the sensor rail is off by default at power-up, in the same way the 100 kΩ on the boost EN is needed for the pump.

Configuration per channel:

- **CT: leave open.** 120 µs turn-on at 3.6 V. Inrush into the HW-390's ~100 nF at that slew rate is only a few milliamps, so there is nothing to limit. Add a capacitor from CT to VIN only if the cable capacitance turns out to matter (3.8 µs/pF at 3.6 V).
- **QOD: tie directly to VOUT.** With the internal ~150 Ω and ~100 nF of load, fall time is roughly 30 µs. Left floating, the output stays floating after shutdown, which is the wrong answer when a metre of cable is attached.
- **CIN: 1 µF at the pin.** TI calls for 1 µF in most situations, with short traces and the caps close to the device.
- **ON: drive it explicitly low before sleep.** Once ON has been driven high the smart pulldown is out of circuit, so a high-Z pin holds its charge rather than defaulting off.

Pinout for footprint work: 1 VIN, 2 GND, 3 ON, 4 CT, 5 QOD, 6 VOUT.

Costs. Three SOT-23-6 packages plus three input capacitors, placed near connectors that are already crowded — fold this into the re-placement in §4.1. And 2 A at 80 mΩ is enormous overkill for 5 mA, so roughly a dollar per channel buys the QOD, the defined-off state and the reverse blocking rather than the current rating.

It will not work on the +9V pump rail. Absolute maximum VIN is 6 V. If you also want a real disconnect for the pump — worth having, since the boost leaves roughly VIN minus a diode drop on `+9V` even when EN is low — you need a part rated well above 9 V with margin. TI's TPS228xx high-voltage load switches or a discrete P-FET with a gate driver cover that range; verify the specific part against 9 V before committing.

The boost EN pulldown from §3.2 is still required either way. A load switch elsewhere does not address it.

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6. Should this be a 4-layer board?

Yes — but not for routing density.

There are 891 mm of track on F.Cu and 207 mm on B.Cu across a 3884 mm² board. This design is not short of routing space; it could be finished on two layers.

The reason to go to four layers is that a 2.4 GHz radio, a switching converter and four analog sensor channels are all sharing one reference plane that is currently sliced in three places (under the module, by the 22 mm AEM slot, and by the 25 mm battery trace). On four layers L2 is untouched copper and all three problems disappear at once.

For a 76 × 51 mm board in quantity 5, the price difference at JLCPCB or PCBWay is small enough not to deliberate over.

6.1 Recommended stackup

Use the fab's standard 1.6 mm 4-layer stackup rather than a custom one. JLCPCB's `JLC04161H-7628` is representative:

Layer	Material	Thickness	Role
L1 (F.Cu)	copper	1 oz / 35 µm	Signals, power, all components
—	prepreg 7628	0.2104 mm	
L2	copper	0.5 oz / 15.2 µm	**Solid ground. No routing at all.**
—	FR-4 core	1.065 mm	
L3	copper	0.5 oz / 15.2 µm	Power pours (+3V3, +4V2, +9V, +5V), slow signals
—	prepreg 7628	0.2104 mm	
L4 (B.Cu)	copper	1 oz / 35 µm	Remaining signals, ground fill

Total ≈ 1.585 mm, i.e. nominal 1.6 mm.

The 0.21 mm from L1 to L2 is the number that matters. Everything routed on the top layer gets a return path 0.21 mm below it instead of 1.51 mm, which shrinks every loop on the board by a factor of about seven.

6.2 Rules to hold to

- L2 stays continuous. If you want to route one signal on it, don't — that is what L3 and L4 are for.
- Keep the whole boost power loop on L1 with a local ground pour tied to L2 by a via cluster at the IC's GND pad.
- Analog channels (three soil, one pressure) stay on L1 or L4 over solid copper, routed away from the boost and from the pump PWM.
- The antenna keepout applies to all four layers and extends past the module's edges on every side.
- Stitch the perimeter and the area around the module at roughly 5 mm pitch.
- If L3 carries split power pours, keep the splits away from anything crossing on L1 or L4.

6.3 Impedance

Not required for this design. The module has an integrated antenna, so there is no 50 Ω RF trace, and the CH340E is full-speed USB over a ~10 mm run, which tolerates a mismatch.

If you want to do it properly anyway, at 0.21 mm to L2 with $\varepsilon r \approx 4.4$:

- ≈ 0.35 mm trace for 50 Ω single-ended
- ≈ 0.25 mm traces with a 0.14 mm gap for 90 Ω differential

These come from a closed-form microstrip approximation. Run them through the fab's calculator against their actual stackup before committing.

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7. Suggested order of work

1. Pressure sensor divider, boost EN pull-down, CH340E supply gating. Functional or destructive bugs.
2. LDO input capacitor, VBUS bulk, pump connector decoupling, protection on all external lines (§5). Decide on load switches for the three sensor rails at the same time (§5.7), since they change the connector-area footprint.
3. Re-place so the antenna faces clear board with no connectors or cables near it, then enforce the vendor keepout on all layers.
4. Move to 4 layers and rebuild the routing with L2 untouched.
5. Tighten the boost: inductor next to SW, small capacitor across pads 5/6, via cluster at the GND pad, shrink the SW pour.
6. Annotate reference designators and switch to ENIG.

Steps 1 and 2 are schematic work that can be done immediately. Step 4 makes steps 3 and 5 much easier, so sequence the re-layout after the layer decision rather than patching the current 2-layer routing.